

Practical-2

Aim: Hadoop Installation, Configuration and HDFS Operations on Windows/Linux

Problem Definition:

You have recently joined DataAnalytics Inc. as a Hadoop Administrator.

The organization processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to increasing data volumes, the company has decided to migrate its data storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS).

As a Hadoop Administrator, your responsibility is to install and configure Hadoop in a Single-Node Cluster environment on Windows/Linux, verify the Hadoop services, manage HDFS storage, and perform data management operations.

Management wants a fully functional Hadoop environment that can store, retrieve, and process big data efficiently.

Your task is to install Hadoop, configure the Hadoop ecosystem, start Hadoop services, and perform HDFS operations using Hadoop commands.

Task:

- Hadoop Installation & Configuration
- Access Hadoop Web UI.

1. Create directories in HDFS.

```
gitanshu@Parth:~$ hdfs dfs -mkdir /MarketingAnalytics
```

Subdirectories:

```
gitanshu@Parth:~$ hdfs dfs -mkdir /MarketingAnalytics/RawData
hdfs dfs -mkdir /MarketingAnalytics/ProcessedData
hdfs dfs -mkdir /MarketingAnalytics/Reports
gitanshu@Parth:~$ hdfs dfs -mkdir /MarketingAnalytics/ProcessedData
gitanshu@Parth:~$ hdfs dfs -mkdir /MarketingAnalytics/Reports
```

2. Upload files to HDFS.

```
gitanshu@Parth:~$ nano sales_data.txt
gitanshu@Parth:~$ hdfs dfs -put sales_data.txt /MarketingAnalytics/RawData
```

3. List HDFS files and directories.

```
gitanshu@Parth:~$ hdfs dfs -ls /
Found 1 items
drwxr-xr-x - gitanshu supergroup          0 2026-08-01 08:14 /MarketingAnalytics
gitanshu@Parth:~$ hdfs dfs -ls /MarketingAnalytics
Found 3 items
drwxr-xr-x - gitanshu supergroup          0 2026-08-01 08:14 /MarketingAnalytics/ProcessedData
drwxr-xr-x - gitanshu supergroup          0 2026-08-01 08:14 /MarketingAnalytics/RawData
drwxr-xr-x - gitanshu supergroup          0 2026-08-01 08:14 /MarketingAnalytics/Reports
gitanshu@Parth:~$ hdfs dfs -ls /MarketingAnalytics/RawData
Found 1 items
-rw-r--r--  1 gitanshu supergroup          63 2026-08-01 08:14 /MarketingAnalytics/RawData/sales_data.txt
```

4. Display file contents.

```
gitanshu@Parth:~$ hdfs dfs -cat /MarketingAnalytics/RawData/sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 200
```

5. Copy files within HDFS.

```
gitanshu@Parth:~$ hdfs dfs -cp /MarketingAnalytics/RawData/sales_data.txt /MarketingAnalytics/Reports
```

6. Move files within HDFS.

```
gitanshu@Parth:~$ hdfs dfs -mv /MarketingAnalytics/Reports/sales_data.txt /MarketingAnalytics/ProcessedData
```

7. Rename files.

```
gitanshu@Parth:~$ hdfs dfs -mv /MarketingAnalytics/ProcessedData/sales_data.txt /MarketingAnalytics/ProcessedData/final_sales.txt
```

8. Download files from HDFS.

```
gitanshu@Parth:~$ mkdir Download
hdfs dfs -get /MarketingAnalytics/ProcessedData/final_sales.txt Download/
gitanshu@Parth:~$
```

9. Delete files and directories.

```
gitanshu@Parth:~$ hdfs dfs -rm /MarketingAnalytics/ProcessedData/final_sales.txt
Deleted /MarketingAnalytics/ProcessedData/final_sales.txt
gitanshu@Parth:~$ hdfs dfs -rm -r /MarketingAnalytics
Deleted /MarketingAnalytics
```

10. Check HDFS report.

```
gitanshu@Parth:~$ hdfs dfsadmin -report
Configured Capacity: 269490393088 (250.98 GB)
Present Capacity: 251655016448 (234.37 GB)
DFS Remaining: 251654987776 (234.37 GB)
DFS Used: 28672 (28 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Badly Distributed Blocks: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Badly Distributed Blocks: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: Parth.localdomain
Decommission Status : Normal
Configured Capacity: 269490393088 (250.98 GB)
DFS Used: 28672 (28 KB)
Non DFS Used: 4074704896 (3.79 GB)
DFS Remaining: 251654987776 (234.37 GB)
DFS Used%: 0.00%
DFS Remaining%: 93.38%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sat Aug 01 08:18:01 UTC 2026
Last Block Report: Sat Aug 01 05:45:25 UTC 2026
Num of Blocks: 0
```

11. Verify block information.

```
gitanshu@Parth:~$ hdfs fsck /
Connecting to namenode via http://localhost:9870/fsck?ugi=gitanshu&path=%2F
FSCK started by gitanshu (auth:SIMPLE) from /127.0.0.1 for path / at Sat Aug 01 08:19:11 UTC 2026

Status: HEALTHY
  Number of data-nodes: 1
  Number of racks:      1
  Total dirs:           1
  Total symlinks:       0

Replicated Blocks:
  Total size:    0 B
  Total files:   0
  Total blocks (validated): 0
  Minimally replicated blocks: 0
  Over-replicated blocks: 0
  Under-replicated blocks: 0
  Mis-replicated blocks: 0
  Default replication factor: 1
  Average block replication: 0.0
  Missing blocks: 0
  Corrupt blocks: 0
  Missing replicas: 0
  Blocks queued for replication: 0

Erasure Coded Block Groups:
  Total size:    0 B
  Total files:   0
  Total block groups (validated): 0
  Minimally erasure-coded block groups: 0
  Over-erasure-coded block groups: 0
  Under-erasure-coded block groups: 0
  Unsatisfactory placement block groups: 0
  Average block group size: 0.0
  Missing block groups: 0
  Corrupt block groups: 0
  Missing internal blocks: 0
  Blocks queued for replication: 0
FSCK ended at Sat Aug 01 08:19:11 UTC 2026 in 19 milliseconds
```

12. Check Hadoop cluster health.

```
gitanshu@Parth:~$ hdfs dfsadmin -report
Configured Capacity: 269490393088 (250.98 GB)
Present Capacity: 251655016448 (234.37 GB)
DFS Remaining: 251654987776 (234.37 GB)
DFS Used: 28672 (28 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Badly Distributed Blocks: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Badly Distributed Blocks: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: Parth.localdomain
Decommission Status : Normal
Configured Capacity: 269490393088 (250.98 GB)
DFS Used: 28672 (28 KB)
Non DFS Used: 4074704896 (3.79 GB)
DFS Remaining: 251654987776 (234.37 GB)
DFS Used%: 0.00%
DFS Remaining%: 93.38%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sat Aug 01 08:19:31 UTC 2026
Last Block Report: Sat Aug 01 05:45:25 UTC 2026
Num of Blocks: 0
```

13. Monitor Hadoop processes.

```
gitanshu@Parth:~$ jps
8754 Jps
5797 NameNode
5926 DataNode
407 ResourceManager
1577 NodeManager
6159 SecondaryNameNode
```

14. Check storage utilization.

```
gitanshu@Parth:~$ hdfs dfs -du -h /
gitanshu@Parth:~$
```

Key Questions /Analysis:

1. Why is Hadoop preferred for Big Data storage?

Hadoop is preferred for Big Data storage because it can store and process very large amounts of data efficiently. It uses HDFS to distribute data across multiple machines, providing scalability and fault tolerance.

Hadoop also supports parallel processing, making data analysis faster. It is cost-effective as it runs on commodity hardware. Therefore, it is widely used for Big Data applications.

2. What is the role of NameNode and DataNode in HDFS? The

NameNode is the master node that manages the metadata and directory structure of HDFS. It keeps track of file locations and block information.

The DataNode stores the actual data blocks and performs read/write operations. It regularly sends heartbeat signals to the NameNode.

Together, they ensure efficient and reliable data storage.

3. What is the difference between HDFS and traditional file systems?

HDFS is a distributed file system designed for storing Big Data across multiple machines, while traditional file systems store data on a single machine. HDFS provides fault tolerance through data replication and is highly scalable. Traditional file systems have limited scalability and no automatic replication. HDFS is optimized for parallel processing and large datasets.

4. What are the advantages of distributed storage?

Distributed storage improves data availability by storing data across multiple nodes. It provides fault tolerance through replication and ensures high reliability. It also allows easy scalability by adding new nodes to the cluster. Parallel access to data improves processing speed. Overall, distributed storage is efficient for handling large-scale data.

1. Why are backups important in enterprise environments?

Backups protect important business data from accidental deletion, hardware failure, ransomware attacks, and system crashes. They ensure business continuity and quick data recovery.

Supplementary Problems:

1. Configure Hadoop pseudo-distributed mode.

```
gitanshu@Parth:~$
gitanshu@Parth:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Parth]
gitanshu@Parth:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
gitanshu@Parth:~$ jps
960 ResourceManager
1460 Jps
359 NameNode
728 SecondaryNameNode
489 DataNode
```

2. Change HDFS replication factor and observe results.

```
gitanshu@Parth:~$ nano $HADOOP_HOME/etc/hadoop/hdfs-site.xml
gitanshu@Parth:~$ echo "This is Hadoop Replication Test" > sample.txt
gitanshu@Parth:~$ hdfs dfs -mkdir -p /Practical2
gitanshu@Parth:~$ hdfs dfs -put sample.txt /Practical2/
gitanshu@Parth:~$ hdfs fsck /Practical2/sample.txt -files -blocks -locations
Connecting to namenode via http://localhost:9870/fsckugi=gitanshu@files-1&blocks=1&locations=1&path=32fPractical2&fsample.txt
FSCK started by gitanshu (auth:SIMPLE) from /127.0.0.1 for path /Practical2/sample.txt at Fri Aug 07 09:25:08 UTC 2026

/Practical2/sample.txt 32 bytes, replicated: replication=2, 1 block(s): Under-replicated BP-1044796463-127.0.1.1-1785563106885:blk_1073741827_1003. Target Replicas is 2 but found 1 live replica(s), 0 decommissioned replica(s), 0 decommissioning replica(s).
0, BP-1044796463-127.0.1.1-1785563106885:blk_1073741827_1003 len=32 Live_repl=1 [DataNodeInfoWithStorage[127.0.0.1:9866,DS-f2874b82-d4fa-4ec0-88f2-92f0825028fb,DISK]]

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 0
Total symlinks: 0

Replicated Blocks:
Total size: 32 B
Total files: 1
Total blocks (validated): 1 (avg. block size 32 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 1 (100.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 1 (50.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Aug 07 09:25:08 UTC 2026 in 26 milliseconds
```

3. Upload a large dataset into HDFS.

```
The filesystem under path '/Practical2/sample.txt' is HEALTHY
gitanshu@Parth:~$ hdfs dfs -mkdir -p /datasets
gitanshu@Parth:~$ hdfs dfs -put ~/Downloads/diabetes.csv /datasets/
put: `/home/gitanshu/Downloads/diabetes.csv': No such file or directory
gitanshu@Parth:~$ hdfs dfs -ls /datasets
gitanshu@Parth:~$
```

4. Analyze HDFS storage utilization.

```
gitanshu@Parth:~$ hdfs dfs -du -h /
32 64 /Practical2
0 0 /datasets
gitanshu@Parth:~$ hdfs dfsadmin -report
Configured Capacity: 269490393088 (250.98 GB)
Present Capacity: 251588452395 (234.31 GB)
DFS Remaining: 251588415488 (234.31 GB)
DFS Used: 36907 (36.04 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 1
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Badly Distributed Blocks: 1
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Badly Distributed Blocks: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: Parth.localdomain
Decommission Status : Normal
Configured Capacity: 269490393088 (250.98 GB)
DFS Used: 36907 (36.04 KB)
Non DFS Used: 4141268949 (3.86 GB)
DFS Remaining: 251588415488 (234.31 GB)
DFS Used%: 0.00%
DFS Remaining%: 93.36%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Aug 07 09:29:02 UTC 2026
Last Block Report: Fri Aug 07 09:28:32 UTC 2026
Num of Blocks: 1
```



```

gitanshu@Parth:~$ hdfs fsck /
Connecting to namenode via http://localhost:9870/fsck?ugi=gitanshu&path=%2F
FSCK started by gitanshu (auth:SIMPLE) from /127.0.0.1 for path / at Fri Aug 07 09:29:39 UTC 2026

/Practical2/sample.txt: Under replicated BP-1044796463-127.0.1.1-1785563106885:blk_1073741827_1003. Target Replicas is 2 but found 1 live replica(s), 0 decommissioned replica(s), 0 decommissioning replica(s)

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 3
Total symlinks: 0

Replicated Blocks:
Total size: 32 B
Total files: 1
Total blocks (validated): 1 (avg. block size 32 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 1 (100.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 1 (50.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Aug 07 09:29:39 UTC 2026 in 3 milliseconds

The filesystem under path '/' is HEALTHY
gitanshu@Parth:~$ hdfs fsck /datasets/diabetes.csv -files -blocks -locations
Filesystem is inaccessible due to:
java.io.FileNotFoundException: File does not exist: /datasets/diabetes.csv

```

```

gitanshu@Parth:~$ jps
960 ResourceManager
3298 Jps
359 NameNode
728 SecondaryNameNode
489 DataNode
1087 NodeManager
gitanshu@Parth:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb         251G   3.9G  235G   2% /
none            1.9G   4.0K   1.9G   1% /mnt/wsl
tools           238G  109G  129G  46% /init
none            1.9G   12K   1.9G   1% /run
none            1.9G    0    1.9G   0% /run/lock
none            1.9G    0    1.9G   0% /run/shm
none            1.9G    0    1.9G   0% /run/user
tmpfs           1.9G    0    1.9G   0% /sys/fs/cgroup
drivers          238G  109G  129G  46% /usr/lib/wsl/drivers
lib             238G  109G  129G  46% /usr/lib/wsl/lib
drvfs           238G  109G  129G  46% /mnt/c
gitanshu@Parth:~$

```

Tools /Technology used:

- UbuntuLinux/Kali Linux (WSL)
- Hadoop3.x
- OpenJDK11orabove
- HDFS(HadoopDistributed File System)
- LinuxTerminal
- BashShell
- SSH(OpenSSHServer)
- HadoopWebUI(NameNode–Port9870)

TextEditor (Nano / Vim)

Conclusion :

The Hadoop single-node cluster was successfully installed and configured on Linux. HDFS services were started, and various file management operations such as creating directories, uploading, downloading, copying, moving, renaming, and deleting files were performed successfully. The Hadoop Web UI and HDFS reports were used to verify cluster status, storage utilization, and DataNode information. This practical provided hands-on experience in Hadoop administration and demonstrated the efficient management of distributed storage using HDFS.

Github Repo:

https://github.com/Gitanshu1504/24DIT070_BDA_Practicals_2026_SHP